

Assignment 1 (25%)
STQD6324 Data Management
SEMESTER 2 2025/2026

In this assignment, you are required to use **Spark MLlib** to perform a **classification task** on the `Iris` dataset. The dataset can be obtained from publicly available online sources.

You may follow the steps below as a general guideline:

- Load the `Iris` dataset into a **Spark DataFrame**.
- Perform necessary **data preprocessing** (if applicable).
- Split the dataset into **training and testing sets**.
- Select and implement **three classification algorithms** (e.g., Decision Tree, Random Forest, or Logistic Regression) using Spark MLlib.
- Apply **model tuning techniques**, such as cross-validation and grid search, to optimise the hyperparameters of each selected model.
- Evaluate the performance of the tuned models using appropriate evaluation metrics (e.g., accuracy, precision, recall, F1-score).
- Generate predictions on the testing dataset using the optimised model.
- Conduct a **comparative analysis** across the three models, including:
 - Performance comparison using evaluation metrics
 - Strengths and limitations of each model
 - Justification of the best-performing model

You are encouraged to explore **creative approaches**, justify your methodological choices, and leverage online resources where appropriate.

For each steps performed, provide **clear explanations and interpretations** to guide readers in understanding your workflow, decisions, and findings.

Expected Deliverables

You are required to submit the following via a **GitHub repository**:

- 1. Jupyter Notebook / PySpark Notebook**
 - Complete implementation of the workflow
 - Well-commented code
 - Clear explanations and interpretations for each step
- 2. README.md**
 - Overview of the project
 - Description of dataset and methodology
 - Summary of results and key findings
 - Instructions to reproduce the analysis

Submission Instructions:

Please submit your completed work via GitHub by **2026-05-31**.

Assessment Rubric (Total: 25 Marks)

Criteria	Excellent	Intermediate	Poor
	7-8	4-6	0-3
Technical Implementation (8 marks)	All three models are correctly and fully implemented using Spark MLlib; proper data handling; code runs without errors; demonstrates strong understanding	Most implementations are correct; minor issues in code or execution; one model may be weakly implemented	Incomplete or incorrect implementation; major errors; fewer than three models implemented
	4-5	2-3	0-1
Model Tuning & Evaluation (5 marks)	Systematic use of cross-validation and grid search; appropriate metrics selected and clearly justified	Basic tuning applied; limited parameter search; metrics used but not fully justified	Minimal or no tuning; inappropriate or missing evaluation metrics
	4-5	2-3	0-1
Comparative Analysis (5 marks)	Insightful comparison across all three models; clear justification of best model; discusses strengths and limitations	Basic comparison provided; limited discussion; weak justification	Little or no comparison; lacks critical analysis
	3-4	2	0-1
Interpretation & Explanation (4 marks)	Clear, logical explanations; strong interpretation of results; demonstrates deep understanding	Some explanation provided but lacks clarity or depth	Minimal or unclear explanation; poor interpretation
	3	2	0-1
Code Quality & Reproducibility (3 marks)	Well-organised code; clear comments; proper GitHub structure; fully reproducible	Code is readable but lacks consistency or documentation; partially reproducible	Poor structure; minimal documentation; not reproducible